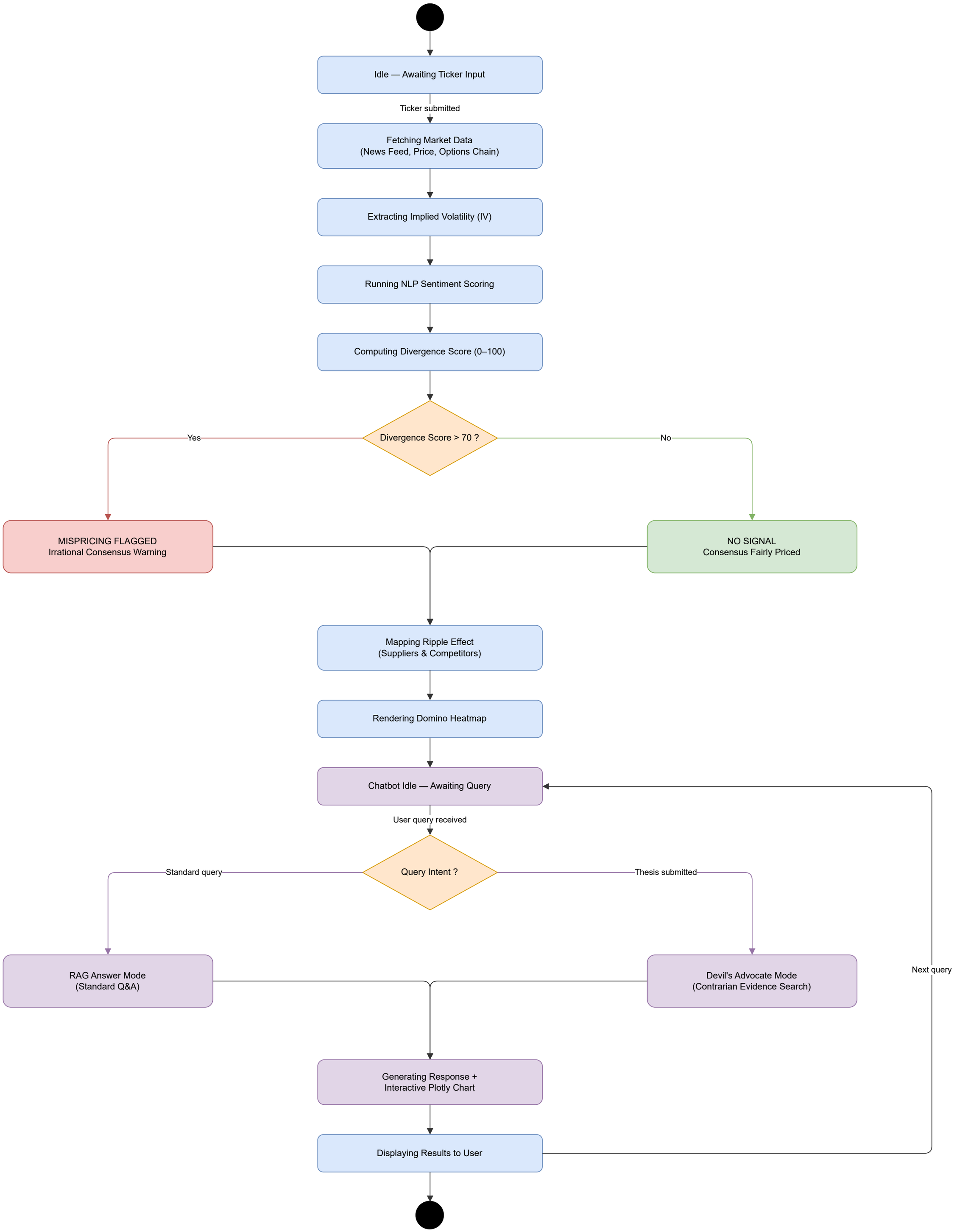
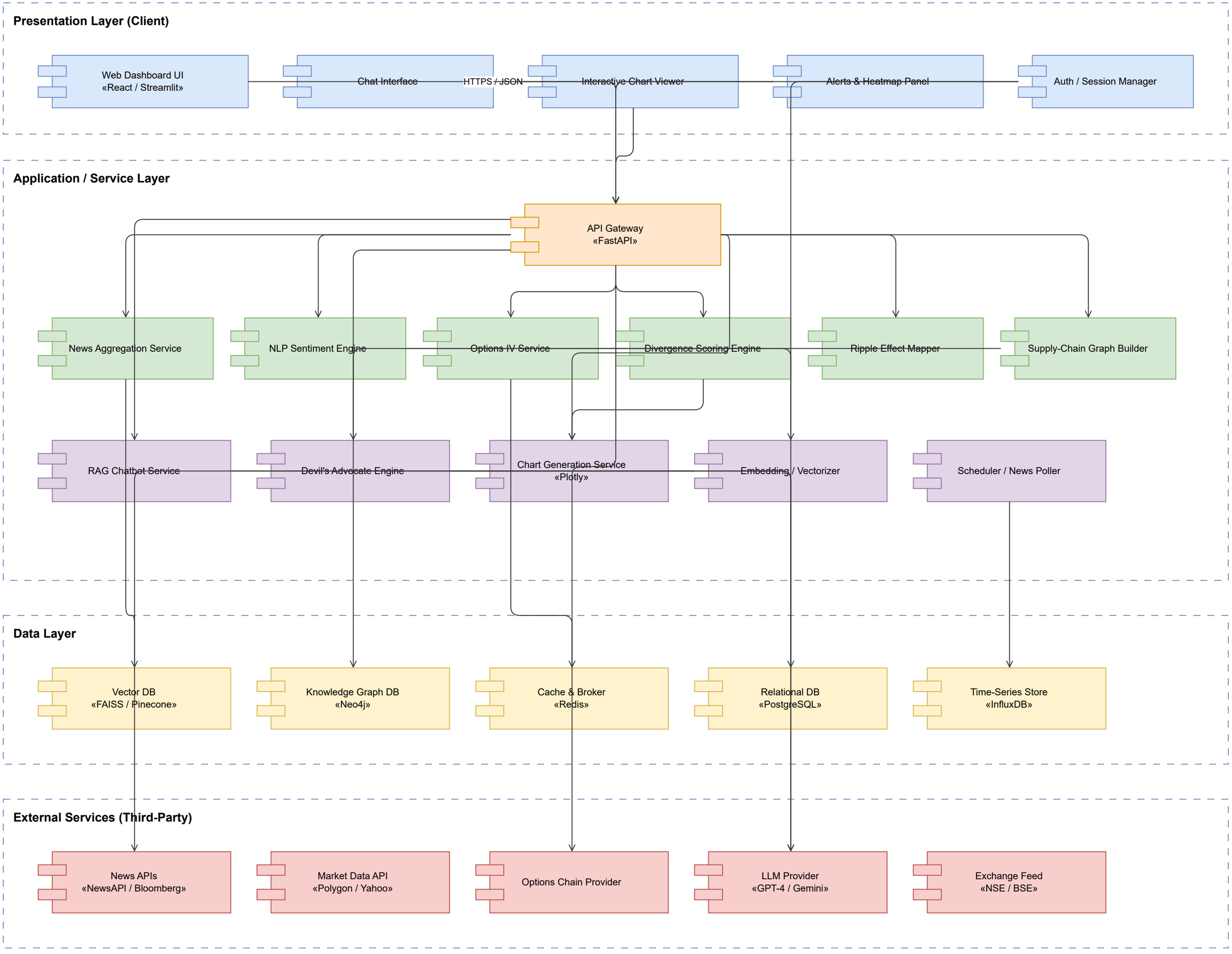
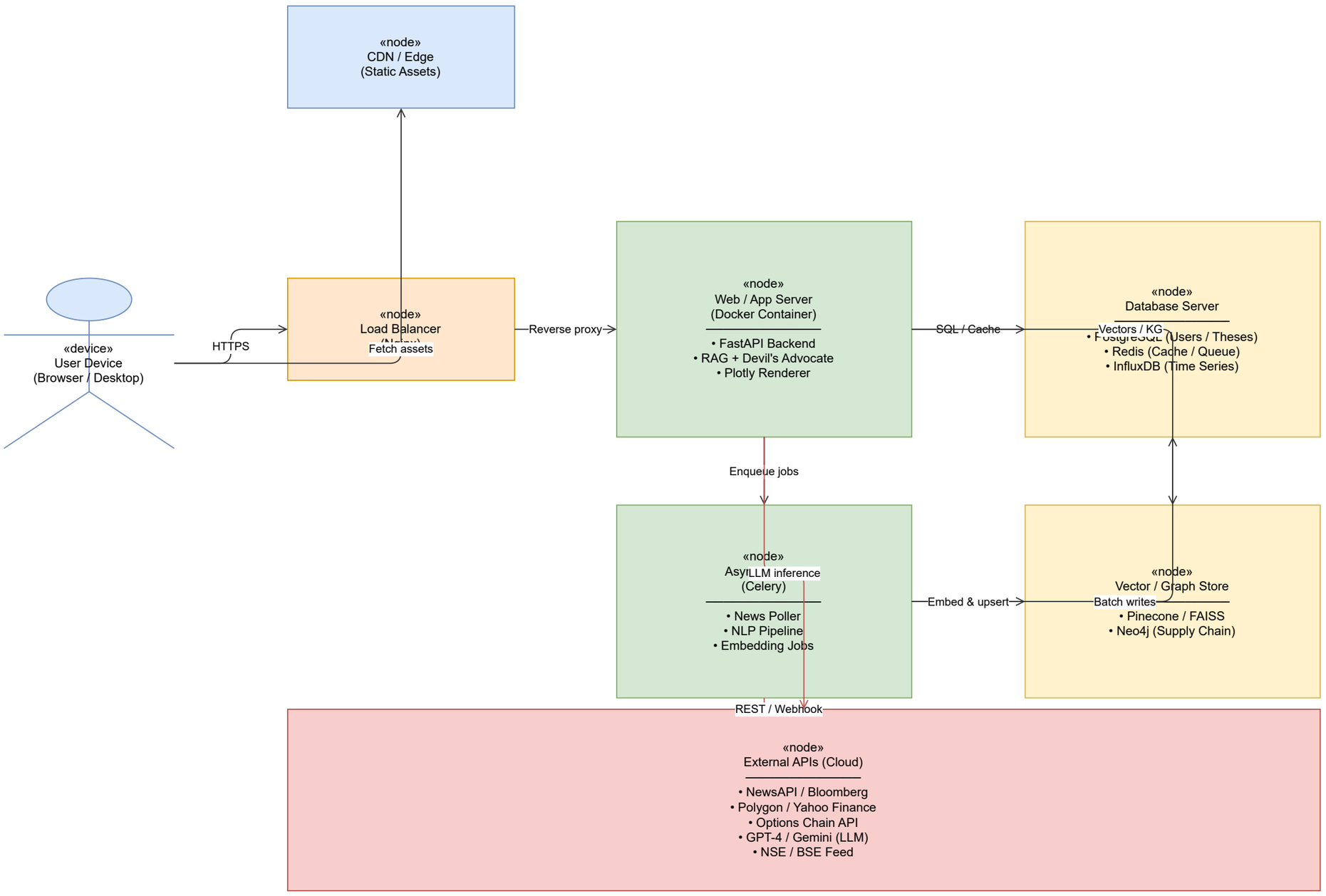

Idle — Awaiting Ticker Input
Ticker submitted
Fetching Market Data
(News Feed, Price, Options Chain)
Extracting Implied Volatility (IV)
Running NLP Sentiment Scoring
Computing Divergence Score (0–100)
Divergence Score > 70 ?
Yes
No
MISPRICING FLAGGED
Irrational Consensus Warning
NO SIGNAL
Consensus Fairly Priced
Mapping Ripple Effect
(Suppliers & Competitors)
Rendering Domino Heatmap
Chatbot Idle — Awaiting Query
User query received
Query Intent ?
Standard query
Thesis submitted
RAG Answer Mode
(Standard Q&A)
Devil's Advocate Mode
(Contrarian Evidence Search)
Generating Response +
Interactive Plotly Chart
Displaying Results to User
Next query


Presentation Layer (Client)
Web Dashboard UI
«React / Streamlit»
Chat Interface
HTTPS / JSON
Interactive Chart Viewer
Alerts & Heatmap Panel
Auth / Session Manager
Application / Service Layer
API Gateway
«FastAPI»
News Aggregation Service
NLP Sentiment Engine
Options IV Service
Divergence Scoring Engine
Ripple Effect Mapper
Supply-Chain Graph Builder
RAG Chatbot Service
Devil's Advocate Engine
Chart Generation Service
«Plotly»
Embedding / Vectorizer
Scheduler / News Poller
Data Layer
Vector DB
«FAISS / Pinecone»
Knowledge Graph DB
«Neo4j»
Cache & Broker
«Redis»
Relational DB
«PostgreSQL»
Time-Series Store
«InfluxDB»
External Services (Third-Party)
News APIs
«NewsAPI / Bloomberg»
Market Data API
«Polygon / Yahoo»
Options Chain Provider
LLM Provider
«GPT-4 / Gemini»
Exchange Feed
«NSE / BSE»


«node»
CDN / Edge
(Static Assets)
«device»
User Device
(Browser / Desktop)
HTTPS
«node»
Load Balancer
(Nginx)
Fetch assets
Reverse proxy
«node»
Web / App Server
(Docker Container)
• FastAPI Backend
• RAG + Devil's Advocate
• Plotly Renderer
SQL / Cache
«node»
Database Server
Vectors / KG
• PostgreSQL (Users / Theses)
• Redis (Cache / Queue)
• InfluxDB (Time Series)
Enqueue jobs
«node»
Async Worker
(Celery)
• News Poller
• NLP Pipeline
• Embedding Jobs
LLM inference
Embed & upsert
«node»
Vector / Graph Store
Batch writes
• Pinecone / FAISS
• Neo4j (Supply Chain)
REST / Webhook
«node»
External APIs (Cloud)
• NewsAPI / Bloomberg
• Polygon / Yahoo Finance
• Options Chain API
• GPT-4 / Gemini (LLM)
• NSE / BSE Feed